

Education

BSc Physics | King's College London

Sep 2016 – May 2020

Publication | [Comparative Study of How Additives Affect Sintered Silver](#) (IMAPS HiTEN 2019)

July 2019

Professional Experience

AI Strategy & Data Consultant | Self-Employed

Oct 2024 – Present

- Forward-Deployed Engineering: Embedded as an engineer with a West African insurer to collapse fragmented workflows. Engineered a data-extraction and synthesis pipeline that processes unstructured company financials. Compressed manual parsing steps by ~98%.
- Built [Sponge](#): a Graph-RAG platform automating data capture into a 1,000-node Knowledge Graph; engineered a "propose-approve-commit" pipeline for "semantic debugging", allowing for a rigorous HITL reasoning review process and 75% faster investigation of LLM errors/hallucinations.
- Early-stage SaaS startup (Defence sector): digitised the British Army's strategic-planning doctrine, shipping the ingestion layer and seven planning stages across a 42-page React and TypeScript application over 109 PostgreSQL tables and 333 API endpoints.
- Audited all 109 tables across 27 schema modules against 3NF, cleared seven of 14 documented violations through reviewed migrations, and allowed the four downstream stages to receive data
- Agentic Automations: Deployed bespoke AI workflows for several SMEs (e.g., Python recipe-scaling for caterers, email triage for a Theatre Group), reducing manual operational latency by 85%.

Data Scientist | Man Group PLC

Sep 2023 – Dec 2024

- Data Lifecycle Management: Managed the end-to-end integration of 60+ alternative datasets (IoT, Spend, Energy), overseeing the journey from vendor negotiation to production-ready ingestion.
- Infrastructure Engineering: Built Snowflake-native SQL optimisation tools, accelerating operations on incoming datasets by 75% and reducing engineering support overhead by 25%.
- Stakeholder Influence: Led weekly "Data Strategy Pitches" to executive investors, distilling complex technical findings into high-conviction narratives for stakeholders of varying technical literacy.
- Designed an $\$n^2$ cost-sharing framework to pool data resources across 5 funds, split via PIT AUM.
- Codified k-NN entity mapping systems in Python to allow a \$20M global systematic equities fund to trade on alternative data with no ticker symbols for the first time.

Quantitative Researcher (Equities) | Man AHL

Sep 2022 – Sep 2023

- Alpha Research and Predictor Implementation: Researched 30+ alternative datasets to engineer and deploy 6 systematic alpha predictors into production across EU, US, and Asian markets.
- \$6M total allocation with a 9% average return profile over a 1yr lookback.
- Process Engineering: Systemised the research-to-production pipeline for consumer alternative data, reducing time to implementation of these predictors by 75%.

Risk Engineer | Man Group PLC

Sep 2021 – Sep 2022

- Productised 'Spotlight', a high-fidelity risk-visualisation platform, by integrating dozens of disparate data sources into a unified analytical dashboard for Portfolio and Risk Managers.
- Iterative Development: Triaged user-reported errors and analysed usage metrics to transition the firm from legacy back-office tools to a modernised, streamlined UI.
- Delivered simplified real-time risk metrics (e.g. the Greeks) across a \$250Bn multi-asset portfolio, resolving technical debt while meeting front-office performance requirements.

Risk Engineer (Intern) | Man Group PLC

June 2021 – Sep 2021

- Modernised legacy 2008-era VBA risk infrastructure into production-grade Python/Airflow pipelines, reducing manual intervention by 90% for high-dimensional sensitivity models.

Additional information

Competencies: Python, SQL, C#, PostgreSQL, Snowflake, Airflow, Git, Graph-RAG, Agentic Orchestration

Languages: English (native), Spanish (conversational)

Interests: Baking and entrepreneurship: founded and operated AK Bakes, a home bakery generating >£2k revenue; Enabling a strategic move toward an AI-driven backend that automates inventory and customer intake.

Website: ak-baking.square.site